

HHP – HULDRA HALEPRODUKSJON

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SYSTEM CONTROL DIAGRAM - HVAC SYSTEM, MODULE SEQUENCES		
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System Control Diagram

HVAC System, Module Sequences

START HVAC, HHP MODULE SEQUENCE

Phase	Step	Actions	Outputs to	Next step	Wait before next step	Jump condition	Jump to step	Remarks
Start permissive	0	NA	NA	1 or 2	System not available: 77-GK81A.XE and 77-GK81B.XE = TRUE OR 77-GK82A.XE and 77-GK82B.XE = TRUE	System available: 77-GK81A.XE or 77-GK81B.XE = FALSE AND 77-GK82A.XE or 77-GK82B.XE = FALSE	2	
System not available	1	1. Warning on OS	1. 77-XA 9201	END	NA	NA	NA	
Set system in auto	2	1. Set system in auto	1. 77-GK81A.LA 2. 77-GK81B.LA 3. 77-GK82A.LA 4 77-GK82B.LA	3	NA	NA	NA	
Open all dampers	3	1. Open all dampers Start timer 10 sec.	2. 1. 77-GE153.XH 2. 77-GS81.XH 3. 77-GS82.XH 4. 77-GS83.XH 5. 77-GS84.XH 6. 77-GS085.XH 7. 77-GS086.XH 8. 77-GS087.XH 9 77-GE158.XH (One signal to F03)	4 or 5	Timer expierd	All dampers open: 77-GE153.BCH and 77-GS81.BCH and 77-GS82.BCH and 77-GS83.BCH and 77-GS84.BCH and 77-GS085.BCH and 77-GS086.BCH and 77-GS087.BCH and 77-GE158.BCH = TRUE	5	
Abort because damper malfunction	4	1. Warning on OS 2. Start HVAC stop sequence	1. 77-XA9202 2. 77-HS9200L	END	NA	NA	NA	
Check duty/standby supply fans	5	NA	NA	6 or 9	NA	77-GK81A.BP1 and 77-GK81B.BP2 = FALSE	9	77-GK81A Duty 77-GK81B Standby
Start 77-GK81A Duty	6	1. Start fan 2. Start timer 8 sec.	1. 77-GK81A.XP1H	7 or 13	Timer expired	Fan running: 77-GK81A.BCH = TRUE	13	
Start 77-GK81B Standby	7	1. Warning on OS 2. Start Standby fan 3. Start timer 8 sec	1. 77-XA9203 2. 77-GK81B-XP2H	8 or 13	Timer expired	Fan running: 77-GK81B.BCH = TRUE	13	
Abort because supply system malfunction	8	1. Warning on OS 2. Start HVAC stop sequence	1. 77-XA9204 2. 77-XA9206 3. 77-HS9200L	END	NA	NA	NA	
Check duty/standby supply fans	9	NA	NA	10 or 23	NA	77-GK81A.BP2 and 77-GK81B.BP1 = FALSE	23	77-GK81A Standby 77-GK81B Duty
Start 77-GK81B Duty	10	1. Start fan 2. Start timer 8 sec.	1. 77-GK81B.XP1H	11 or 13	Timer expired	Fan running: 77-GK81B.BCH = TRUE	13	
Start 77-GK81A Standby	11	1. Warning on OS 2. Start Standby fan 3. Start timer 8 sec	1. 77-XA9203 2. 77-GK81A-XP2H	12 or 13	Timer expired	Fan running: 77-GK81A.BCH = TRUE	13	
Abort because supply system malfunction	12	1. Warning on OS 2. Start HVAC stop sequence	1. 77-XA9204 2. 77-XA9206 3. 77-HS9200L	END	NA	NA	NA	
Check duty/standby extract fans	13	NA	NA	14 or 17	NA	77-GK82A.BP1 and 77-GK82B.BP2 = FALSE	17	77-GK82A Duty 77-GK82B Standby
Start 77-GK82A Duty	14	1. Start fan 2. Start timer 8 sec.	1. 77-GK82A.XP1H	15 or 21	Timer expired	Fan running: 77-GK82A.BCH = TRUE	21	
Start 77-GK82B Standby	15	1. Warning on OS 2. Start Standby fan 3. Start timer 8 sec	1. 77-XA9203 2. 77-GK82B-XP2H	16 or 21	Timer expired	Fan running: 77-GK82B.BCH = TRUE	21	

System Control Diagram

HVAC System, Module Sequences

START HVAC, HHP MODULE SEQUENCE

Phase	Step	Actions	Outputs to	Next step	Wait before next step	Jump condition	Jump to step	Remarks
Abort because extract system malfunction	16	1. Warning on OS 2. Start HVAC stop sequence	1. 77-XA9204 2. 77-XA9207 3. 77-HS9200L	END	NA	NA	NA	
Check duty/standby extract fans	17	NA	NA	18 or 24	NA	77-GK82A.BP2 and 77-GK82B.BP1 = FALSE	24	77-GK82A Standby 77-GK82B Duty
Start 77-GK82B Duty	18	1. Start fan 2. Start timer 8 sec.	1. 77-GK82B.XP1H	19 or 21	Timer expired	Fan running: 77-GK82B.BCH = TRUE	21	
Start 77-GK82A Standby	19	1. Warning on OS 2. Start Standby fan 3. Start timer 8 sec	1. 77-XA9203 2. 77-GK82A-XP2H	20 or 21	Timer expired	Fan running: 77-GK82A.BCH = TRUE	21	
Abort because extraxt system malfunction	20	1. Warning on OS 2. Start HVAC stop sequence	1. 77-XA9204 2. 77-XA9206 3. 77-HS9200L	END	NA	NA	NA	
Activate heater	21	1. Start heater 2. Start heater timeout 8 sec.	1. 77-FE103.XP1H	22 or END	Timer expired	Heater on: 1. 77-FE103.BCH = TRUE	END	
Alarm because heater malfunction	22	1. Warning on OS	1. 77-XA9205	END	NA	NA	NA	
Abort because duty/standby supply system not selected.	23	1. Warning on OS	1. 77-XA9201 2. 77-XA9206	END	NA	NA	NA	
Abort because duty/standby extraxt system not selected.	23	1. Warning on OS 2. Start HVAC stop sequence	1. 77-XA9201 2. 77-XA9206 3. 77-HS9200L	END	NA	NA	NA	
END	END	NA	NA	NA	NA	NA	NA	

System Control Diagram

HVAC system, Module Sequences

STOP HVAC, HHP MODULE SEQUENCE

Phase	Step	Actions	Outputs to	Next step	Wait before next step	Jump condition	Jump to step	Remarks
Start permissive.	0	1. Set all fans and heater to auto.	1. 77-GK81A.LA 2. 77-GK81B.LA 3. 77-GK82A.LA 4. 77-GK82B.LA 5. 77-FE103.LA	1	NA	NA	NA	
Switch of heater	1	1. Stop command to heater	1. 77-FE103.XP1L	2	Heater confirmed of: 1. 77-GE103.BCL=TRUE	NA	NA	
Delay for cool down heater	2	1. Start timer 60 sec.	NA	3	Timer expired	NA	NA	
Stop supply fans	3	1. Stop command to supply fans	1. 77-GK81A.XP1L 2. 77-GK81A.XP2L 3. 77-GK81B.XP1L 4. 77-GK81B.XP2L	4	Fans comfiremd stoped: 77-GK81A.BCL and 77-GK81B.BCL = TRUE	NA	NA	
Stop extract fan	4	1. Stop command to extract fan	1. 77-GK82A.XP1L 2. 77-GK82A.XP2L 3. 77-GK82B.XP1L 4. 77-GK82B.XP2L	5	Fans comfiremd stoped: 77-GK82A.BCL and 77-GK82B.BCL = TRUE	NA	NA	
Close inlet/outlet damper	5	1. Close command to inlet/outlet dampers	1. 77-GE153.XL 2. 77-GE158.XL 3. 77-GS087.XL	END	Dampers confirmed closed: 77-ZSL4088 and 77-ZSL4089 and 77-ZSL4087= TRUE	NA	NA	
END	END	NA	NA	NA	NA	NA	NA	